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What psychological and socio-demographic factors can influence people's intention to use ridesharing during the war? A case study in Ukraine

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ABSTRACT

Public transport services can be disrupted by natural or human-made crises, such as the recent war in Ukraine. Ridesharing has the potential to be used as an alternative to public transport during such crises. However, peoples' attitudes and intentions towards ridesharing during war has not been studied. This study aims to address this critical gap by collecting and analysing travel behaviour data in two Ukrainian cities: Kyiv and Odessa. Exploratory factor analysis identified ten factors influencing ridesharing, namely: *attitudes, perceived behavioural control, subjective norm, ease of use, usefulness, moral norms, trust, perceived safety, emotions,* and *discrimination*. Then, a combined conceptual model based on the Theory of Planned Behaviour, and the Technology Acceptance Model was proposed, to incorporate potential psychological and socio-demographic in the context of a war situation. Structural equation modelling was used to explore the causal relationships between these factors and ridesharing. In the context of war, perceived 'usefulness' affected attitudes, while perceived 'ease of use' influenced perceived behavioural control. Moral norms strongly impacted the ridesharing intention; trust influenced attitudes; and gender played a major role by indirectly affecting ridesharing intention. This can provide transport planners and policy-makers with insights as to how ridesharing can be more attractive and become a genuine tool for enhancing human mobility resilience.

1. Introduction

Passenger and freight transport are the backbones of modern society (Angell & Potoglou, 2022; Nikitas et al., 2019). Efficient and well-planned transport systems enhance the quality of life, promote economic development, and foster connections between individuals and communities, as mentioned by the Sustainable Development Goal (SDG) No. 11 (UN, 2018). Transport resilience, defined as the ability of the transport system to maintain its functions during and after external shocks and disruptions (Liu et al.,

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2024), is thus emerging as a critical concept when looking to alleviate the adverse consequences of these crises. Every city needs a transport network that can continue to provide an accessible and reasonably frequent service across all neighbourhoods, even when public transport links, which are traditionally 'the epicentre of every mobility system', are degraded by disasters (Yonson et al., 2020). Resilient transport systems, that can handle emergency situations, ensuring the swift movement of people and goods including the safe evacuation of civilians from danger-zones to safe areas (Bakhshian et al., 2023), therefore, need sometimes to go above and beyond public transport solutions. This is because mass transit can be disrupted by the crisis, and the fixed scheduling may not be responsive enough to cope with immediate evacuation needs.

Research has shown that shared mobility modes, themselves or in combination with other services such as public transport modes, have the potential to improve the efficiency and sustainability of transport networks (Shaheen et al., 2016; Wallgren et al., 2023). Emerging shared transport systems should also be socially sustainable i.e. accessible and inclusive for all to meet vulnerable social groups' travel needs (Dadashzadeh et al., 2022a; Dadashzadeh et al., 2022b; Dadashzadeh et al., 2024). However, the resilience of shared mobility modes during a crisis should also be considered. Shokouhyar et al. (2021) presented the prospects and obstacles related to the impact of the COVID-19 outbreak on the sustainability of shared mobility, taking into account social, economic, and environmental dimensions. In particular, favourable aspects included: a) an increased significance of social attitudes and cultural factors; b) a reduction in unnecessary rides; and c) a surge in the adoption of environmentally friendly mobility options. On the other hand key challenges emerging were: a) a lack of trust among passengers; b) decreased expenditure by riders on transport; and c) the trivialisation of environmental concerns.

As a shared mobility mode, ridesharing involves connecting passengers and drivers with similar travel routes to share a single vehicle, which can lead to reduced costs, decreased traffic congestion, and environmental benefits (Caulfield, 2009; Nikitas, 2024; Rijavec et al., 2020; Mouratidis, 2022). Ridesharing encompasses various classifications based on the participants' relationships and trip purposes (Pigalle & Aguiléra, 2023), namely: fampools, which involve ridesharing among family and friends; carpooling, where rides are shared for commutes between home and work, often organised by employers; informal ridesharing, where neighbours or members of the same association share a ride; and IT-based ridesharing, where drivers and passengers use digital platforms or apps to connect and coordinate rides e.g., BlaBlaCar.

Ridesharing in the form of carpooling is considered important, not only for the purpose of commuting to work, but also for rapid response during natural disasters and pandemics (Terrier and Audrin, 2022; Li et al., 2018; Ho and Lee, 2020). The swift movement of people, materials and resources to aid the affected population is vital during crises (Balcik et al., 2008). During COVID-19, passengers used ridesharing services as protective measures applied by ridesharing companies (Si et al., 2023).

During the recent war in Ukraine (February 2022 to present), electrically fuelled transport systems were disrupted by the attack on power plants by Russia. Buses and trolleybuses in Kyiv and Odessa did not stop working but were delayed and disrupted because of timetable/route changes due to increased/decreased demand. Additionally, the Kyiv metro did not work because the metro stations served as shelters for people but are still used as shelters during air raid alerts. Waller et al. (2023) examined the impact of the ongoing disruption on traffic behaviour in Ukraine in the first month of the war, looking specifically at the interstate highway network and the cities of Kyiv, Kharkiv, Lviv, Dnipro, Odessa, and Mariupol. They employed zone-averaged congestion indices and demand generation metrics, which were applied to assess the travel dynamics in the three cities of Kyiv, Kharkiv, and Mariupol. They showed the usefulness of crowd-sourced travel time data to determine and relate the impacts of war on travel patterns. In Kyiv, Kharkiv, and Mariupol, the invasion events led to increased congestion levels of approximately 50 %, 55 %, and 30 %, respectively. Furthermore, these events resulted in a surge in the demand for travel of around 200 %, 150 %, and 300 % for the respective cities. However, the methodology developed in this work does not take into account possible damage to a part of the road network.

All these disruptions on public transport services and the increased demand of private car use caused several difficulties for the daily travel of people and raised the need for alternative means of transport. Ridesharing emerged in these circumstances. This was the main motivation for examining the use of ridesharing as an alternative transport mode during a crisis that both reduces traffic congestion on the transport network by reducing the number of single occupancy cars, and also addresses part of the increased demand. However, in times of crisis such as periods of war, people's perspectives and beliefs regarding various transport options play a significant role in shaping their mobility patterns and travel decisions. To the best of the authors' knowledge, there is no study that examines people's attitudes and perceptions towards ridesharing during war. To address this gap, this study aims to examine the role of different factors associated with ridesharing in improving transport resilience during such a crisis. The key research questions are as follows:

- What psychological and demographic factors can influence individuals' ridesharing intention during the war?
- Are these factors different compared to other situations?

Henceforth, the paper is structured as follows: Section 2 reviews the existing literature. Section 3 describes the data collection method, case study area, and modelling approach. Section 4 presents the study findings and discusses our key results. Section 5 concludes the study and provides directions for further research.

2. Literature review

A comprehensive literature review was conducted as a benchmark for this study that considered: the effectiveness of ridesharing for improving sustainability and resilience in cities, especially during crises and emergency evacuations; and determinants, influencing factors, attitudes, perceptions and risks towards ridesharing.

2.1. The role of ridesharing in today's city context

The promotion of app-based urban transport services by key stakeholders can influence the travel behaviour of commuters and help them to actively support efforts looking to develop sustainable cities and resilient transport systems (Narayanan & Antoniou, 2022). Shaheen et al. (2018) highlighted that ridesharing, in particular, can provide numerous societal, employer, and individual benefits including: reducing vehicle miles travelled (VMT); fuel consumption, and greenhouse gas (GHG) emissions; decreasing commute times and commuter stress; enhancing access, mobility, and economic opportunities for low-income and minority communities who may be unable to afford a vehicle; and providing cost savings. Rijavec et al. (2020) also found that ridesharing, using organised park and ride sites can reduce the number of single occupancy vehicles on motorways and consequently reduce traffic congestion and its adverse side-effects. Nikitas (2024) reports that the factors determining people's continued use of ride-sharing services are: satisfaction; perceived usefulness; economic benefits; environmental awareness; and platform incentives.

2.2. The role of ridesharing in crises and disasters

There is limited research on ridesharing during times of major disruption in general, and wars in particular. Most of the resilience-related literature, especially pre-COVID, refers to how ridesharing is employed in evacuation analysis (e.g., Borowski & Stathopoulos, 2020; Li et al., 2018; Moug et al., 2023; Tsuchida & Wilshusen, 1991). Some key highlights of this body of work recommend that: different ridesharing benefits apply to different groups of passengers (Tsuchida & Wilshusen, 1991); intermediate trips made by family members and the identification of vulnerable populations can negatively impact evacuation effectiveness (Li et al., 2018); the need for soft strategies such as context-sensitive emergency notifications (Borowski & Stathopoulos, 2020); the use of vehicles with different capacities so as to minimise the evacuation time and find the optimal vehicle placement. The other important area of research around ridesharing in the context of disruption is very recent and refers to COVID-19 studies. Ridesharing played a critical role in the coronavirus era in many cases and was associated with a variety of factors. Ridesharing for instance, was the only mode of travel that enjoyed increased ridership numbers in the Great Walk of Athens' mega-project during and after COVID-19 (Kyriakidis et al., 2023) something showcasing its potential to be a resilient choice in times of disruption. Ivaldi and Palikot, (2023) suggest that decision-makers promoting ridesharing as a solution to intercity travel during crises should account for the impact of the perceived health risks on prices and find ways to ensure access. Zhang and Liu (2022) reported that ambiguity tolerance and environmental concern directly and positively influence potential users' intention toward adopting ridesharing services during COVID. Also, Rahimi et al. (2021) reported that a wide range of factors is influential on the perceived risk of using shared mobility services in general, and ridesharing in particular, including the socio-demographic attributes, built environment settings, and the virus spread.

2.3. Influencing factors, attitudes and barriers to ridesharing

Studies suggest that people's attitudes and perceptions towards ridesharing are influenced by many factors. Malodia and Singla (2016) found that cost savings constitute the most significant factor in encouraging ridesharing. Zhou et al. (2017) found that trust heavily influences the usage of a ridesharing system and identified three factors that shape trust: social proof; social approval; and self-disclosure. Bachmann et al. (2018) found that normative aspects, such as descriptive and personal norms, combined with perceived behavioural control, predicted ridesharing intentions for passengers and drivers. Park, et al. (2018) suggests that increasing ride-sharing usage can be achieved by focusing on individuals who show interest in ridesharing and have specific role (driver or passenger) preferences, along with associated incentives in mind.

Alonso González et al. (2020) found that the willingness to share a ride depends primarily on the time–cost trade-off, rather than on the disutility stemming from pooling rides per se. König and Grippenkoven (2020) assessed travellers' willingness to share rides in autonomous mobility-on-demand systems and found that cost, time, trust, and social norms are all important factors. In Vaclavik et al. (2020) study, information provision was found to be the most significant variable explaining trust in the ridesharing platform, followed by platform safety and security, with pleasant participation, enjoyment, and cost-effectiveness also playing a significant role. Friman et al. (2020) found that ridesharing was not perceived as particularly accessible and that simplicity of travel, population density, years of education, and school and work trips, affected perceptions of accessibility of ridesharing.

Amirkiaee and Evangelopoulos (2018) examined the motives for participation in ridesharing and found that trust, time benefit, and transportation anxiety were strong predictors, with economic benefits being a marginal predictor. Their findings imply that in situations where transport anxiety is high, if people trust the ridesharing service providers and participants, they will have a strong intention to participate in ridesharing. Terrier and Audrin (2022) studied the risks of using ridesharing and fears of COVID-19 and highlighted the importance of trust. Pratt et al. (2019) analysed online commentary about UberPool and Lyft Shared services on Twitter and found that although negative tweets outnumber positive ones, an evident humorous discourse indicates that sharing rides can lead to rich cultural experiences, despite the drawbacks such as increased travel time and sharing space with strangers.

Table 1 provides a synopsis of the existing literature on ridesharing and disruption, organised by regions, and provides descriptions of the methods used (the vast majority were quantitative analyses), and the key results found with an emphasis on the factors that could underpin intentions to use ridesharing.

2.4. Research gap identification

The research gap identified is the lack of investigation into the psychological factors influencing people's intention to use

Table 1
Literature Synopsis.

No	Authors and Year	Region	Country / City Name	Method	Sample Size	Findings (Factors Affecting Ridesharing Intentions)
1	Rozsa et al., 2023	Eastern Europe	Czechia, Slovak Republic	Query method via questionnaire. Descriptive and factor analysis	311	Demographic, financial, economic factors
2	Malichová et al. (2023)	Eastern Europe	Czechia, Finland, Greece, Italy, Slovakia	Regression analysis	609	Mode of transport, trip origin, distance, comfort perceptions, socio-demographic characteristics (gender and age)
3	Perkumienė et al., 2021	Europe, North America	Europe and the US	Statistical analysis	149	General attitudes towards Online Transport-sharing Platform, price, trust, expected risk factors, security, comfort, authenticity, fun, social relations
4	Cylwik et al., 2020	Eastern Europe	Poland	Unified theory of acceptance and use of technology (UTAUT), factor analysis and linear regression using the partial least squares (PLS) method	240	Outcome expectation, hedonic motivation, habit, effort expectation, price value, trust
5	Rijavec et al., 2020	Eastern Europe	Slovenia	A disaggregated four-step transport model, utility theory, and scenario analysis	247	Travel cost, traveling together is more pleasant, environmental pollution, lack of parking lots at the destination, dissatisfaction with public transport, barriers to carpooling
6	Narayanan & Antoniou, 2022	Western Europe	Germany	Combination of a linear regression, Dirichlet regression, and multinomial logit models	287	Socio-demographic characteristics (age, employment, income, education, household vehicle ownership), use of conventional modes (bicycle, PT, and private car) and supply parameter (fleet size). Operational and policy measures are suggested under the following categories: (i) Mobility as a Service (MaaS), (ii) Pricing strategy, (iii) Hybrid fleet, and (iv) Campaigns.
7	Alonso González et al., 2020	Western Europe	Netherlands	Stated preference (SP) study with discrete choice mixed logit model	1006	The time–cost trade-off is the main factor influencing the percentage of people who are willing to share their rides. Having one or two extra passengers has a constant disutility regardless of the duration of the trip, but if one is riding with four other people, the disutility grows with the distance of the trip.
8	König & Grippenkoven, 2020	Western Europe	Germany	Binary logistic regression	150	Discounts needed to attract shared rides depend on travel time and detour factors, willingness to share rides (willingness to accept), refusal rate, travel time, detour factor
9	Bachmann et al., 2018	Western Europe	Switzerland	The theory of planned behaviour, the norm activation model, dispositional trust.	342	Normative aspects, such as descriptive and personal norms, combined with perceived behavioural control, predicted ridesharing intentions for passengers and drivers. Attitudes towards ridesharing behaviour did not have any predictive power regarding ridesharing intention.
10	Ivaldi & Palikot, 2023	Western Europe	France BlaBlaCar data	Regression analysis	5,602trips	In the areas and periods of increased health risks during COVID-19 pandemic, the number of available city-to-city ridesharing drivers declines, and the prices they charge for their services increase.
11	Kyriakidis et al., 2023	Southern Europe	Greece	Local road user e-survey analysis, twin statistical analysis	385	Ridesharing saw increased ridership during and after COVID-19, indicating resilience in disruptions. Public acceptance, closed space with many others
12	Friman et al., 2020	Northern Europe	Sweden	Multiple linear hierarchical regression analyses	122	Perceived accessibility of ridesharing is low, influenced by population density

(*continued on next page*)

Table 1 (*continued*)

No	Authors and Year	Region	Country / City Name	Method	Sample Size	Findings (Factors Affecting Ridesharing Intentions)
						and travel simplicity. Perceived accessibility, simplicity of travel, population density, years of education, school and work-trips, affect perceptions, travel time, cost.
13	Chen & Lu, 2021	East Asia	China	Protection motivation theory (PMT), Usage situation theory, Explanatory factor analysis (EFA)	346	Security, protective behaviour, protection motivation, threat vulnerability, threat severity, response efficacy, self-efficacy, response cost
14	Wang et al., 2020	East Asia	China, six universities	Extended technology acceptance model (TAM)	426	Personal innovativeness, environmental awareness, perceived risk, perceived usefulness, perceived ease of use
15	Si et al., 2022	East Asia	China	Partial least squares structural equation modelling (PLS-SEM)	614	Satisfaction, perceived usefulness, economic benefits, environmental awareness, social experience, platform incentives
16	Jiang & Lian, 2024	East Asia	China	TAM	951	Perceived usefulness, perceived ease of use, perceived resource scarcity, perceived risks, attitude, behavioural intention
17	Li et al., 2018	East Asia	China	Driver and experts interviews and thematic analysis	1169	Identifies challenges in evacuation effectiveness using shared-mobility services.
18	Akbari et al., 2021	West Asia	Iran	Extended TAM with the Theory of Planned Behaviour (TPB), Structural equation modelling analysis using PLS software	318	The effect of perceived usefulness, subjective norms, satisfaction, behavioural intention, price value
19	Ashrafi et al., 2021	South Asia	Bangladesh	PLS-SEM	388	Perceived value, risk perception
20	Maruf et al., 2021	South Asia	Bangladesh	SEM, EFA, CFA	237	Lack of security and privacy, various technological challenges, and inconvenient transaction systems, perceived ease of use, perceived usefulness, attitude towards using ride-sharing apps
21	Rathnasiri et al., 2024	South Asia	Sri Lanka	TPB, multiple regression analysis	286	Subjective norms, perceived behavioural control, awareness of environmental consequences
22	Malodia & Singla, 2016	South Asia	India	Factor analysis, Statistical Analysis	106	Cost savings are a significant motivator for ridesharing. Other factors are time–convenience, private–public cost, extra travel time, walking time to reach meeting point, waiting time at pickup point, and cost savings
23	Pandita et al., 2021	South Asia	India	The UTAUT model, SEM	287	Performance expectancy, effort expectancy, social influence, facilitating conditions, behavioural intentions
24	Shah, Varghese, Jana, & Mathew, 2020	South Asia	India	SEM	217	Socio-economic status, spending propensity, number of companions, comfort in a ride, security, waiting time
25	Tjokrosaputro (2023)	Southeast Asia	Indonesia	SEM, TPB, TAM	208	Perceived usefulness, perceived ease-of-use, word-of-mouth
26	Le Van et al., 2022	Southeast Asia	Vietnam	Ordered probit models	996	Economic benefit, time preference, enjoyment, demographic influence
27	Tsuchida and Wilshusen, 1991	North America	US	Statistical analysis	587 & 187	Ridesharing usage increased post-earthquake; benefits vary among different passenger groups.
28	Borowski and Stathopoulos, 2020	North America	US	SP survey with random parameter discrete choice model	185	Recommends soft strategies for evacuation services; contextual emergency notifications needed. Determinants of on-demand ridesourcing: interactions between race, gender, and income, and external contextual factors such as the evacuation notification source, consequence severity, immediacy, evacuation distance, unfamiliarity of

(*continued on next page*)

Table 1 (*continued*)

No	Authors and Year	Region	Country / City Name	Method	Sample Size	Findings (Factors Affecting Ridesharing Intentions)
						surroundings, and traveling with others.
29	Zhou et al., 2017	North America	US, Instagram	EFA, logistic regression model	100	Trust influences ridesharing usage: identified factors include social proof and self-disclosure.
30	Park et al., 2018	North America	US	Binary probit models, and a seemingly unrelated bivariate probit regression model	2,527	Ridesharing can increase by targeting interested individuals and providing specific incentives. Age, marital status, university affiliation, commute distance, modal split, and personal attitudes were key factors associated with carpool interest and choice.
31	Terrier & Audrin, 2022	North America	US	Hierarchical regressions	299	Quality of relationships and trust in organisations significantly influence ridesharing. Other factors are perceived risks, organizational identification, attitude, trust, participants' intention
32	Rahimi et al., 2021	North America	US	Bivariate ordered probit modelling approach	915	Personal risk perceptions playing a major role in how individuals chose their mode of transport during the pandemic. Socio-demographic attributes and the spread of the virus significantly influenced perceived risk, suggesting that mitigation strategies should be adaptive to spatial variations.
33	Amirkiaee & Evangelopoulos, 2018	North America	US	TPB	300	Findings imply that in situations where transport anxiety is high, if people trust the ridesharing service providers and participants, they will have a strong intention to participate in ridesharing. Trust, time benefit, transportation anxiety, economic benefits.
34	Pratt et al., 2019	North America	Online commentary about UberPool and Lyft Line (Twitter)	Social media (text mining) analysis (Twitter)	2000 "tweets"	Users often tweet humorously about their experiences, particularly about other passengers. Although negative tweets outnumber positive ones, the humorous discourse indicates that sharing rides can lead to rich cultural experiences, despite the drawbacks like increased travel time and sharing space with strangers.

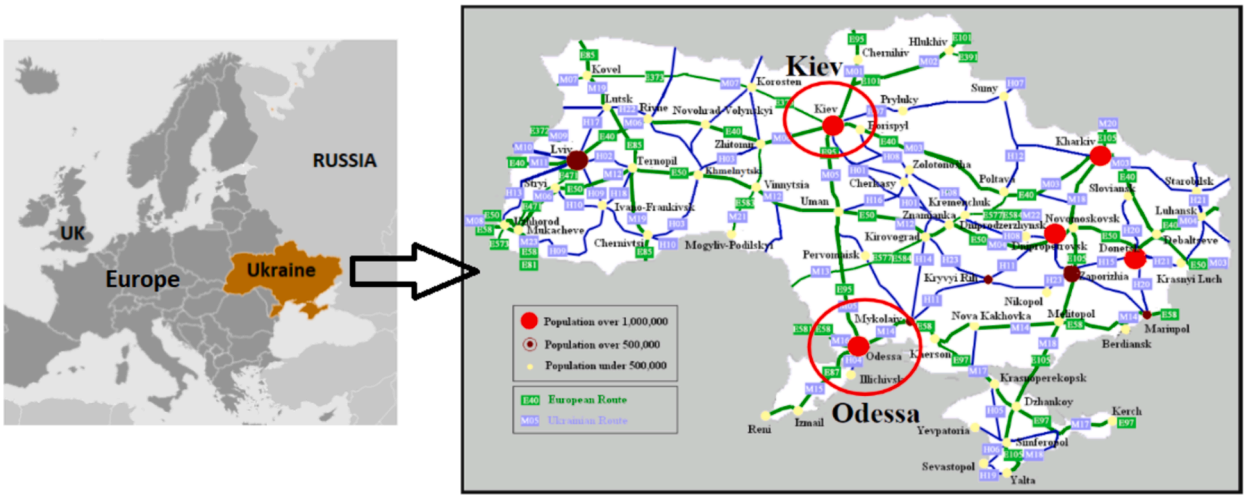


Fig. 1. Location of Ukraine in Europe, Ukraine road map, and the location of Kyiv and Odessa (Highwaysmapseu/Ukraine).

ridesharing during war or crisis situations, particularly within the context of Ukraine's ongoing war. While previous studies have explored ridesharing during natural disasters and health crises (such as earthquakes and COVID-19), no research has specifically examined how people's attitudes, beliefs, and psychological factors impact their decision to use ridesharing during a war. This study addresses this gap by identifying key psychological factors, such as attitudes, perceived behavioural control, subjective norms, and moral norms, and proposing a model that extends traditional frameworks of the Theory of Planned Behaviour (TPB) and Technology Acceptance Model (TAM) to incorporate the unique conditions and challenges of a war environment.

3. Data and method

3.1. Case study: The war in Ukraine

The war in Ukraine commenced on February 24, 2022. Incidences of Russian military actions were observed in major cities across Ukraine, including Berdyansk, Chernihiv, Kharkiv, Odessa, and Kyiv (UN, 2023). The war is still ongoing (at the writing time of this paper, November 2024) and there is still risk of attacks in many cities. Among the attacked areas, two major cities in Ukraine; Kyiv (the capital), and Odessa (one of the most important port cities in Ukraine) were employed as case studies and data collection areas, as shown in Fig. 1. In 2022, the estimated population of Kyiv and Odessa was 2,952,301 and 2,351,392 respectively (Citypopulation.de, 2023a; Citypopulation.de, 2023b).

Kyiv: Infantry battles were fought only in the vicinity of Kyiv (Bucha, Irpin), and not in the city itself. There are different means of transport in Kyiv including metro, tram, trolleybus, bus, minibus (marshrutka), taxi (e.g., Taxi 838), ride hailing (Bolt, Uber, Uklon), bike sharing (Nextbike, BikeNow), and scooter sharing (BikeNow, Bolt, Jet, Kiwi, Zelectra). Before the attacks on the energy infrastructure, there were no problems with electricity and electric transport. The metro did not operate between 24/2/2023 and 17/03/2023 as metro stations were used for shelter during air attacks.

Odessa: Urban transport provision in Odessa is the same as Kyiv with the exception of metro services. Odessa was under missile attacks only. In general, these air attacks still occur with unpredictable frequency. Curfews were introduced, and initially the public transport schedule was changed to match these curfews. Since October 2022, the energy infrastructure of the city has been attacked with missiles. These attacks have caused power outages throughout Ukraine as the power system is highly connected.

In the early days of the war, many tried to flee, which caused delays in the public transport schedule, but trams, buses and trolleybuses did not stop working in Kyiv and Odessa.

Considering these disruptions in public transport services in Kiev and Odessa, as well as local authors' observations, the general psychological state of the local residents can be classified into three stages:

- "Before the Psychological shock-state": Before the air strikes of 24 February 2022.
- "Psychological shock-state": Late February – Mid March 2022 (2–3 weeks)
- "After the Psychological shock-state". Mid March 2022 – Onwards

Although participants were asked about their travel habits before the war (before the psychological shock-state), the main focus of this study is attitudes and perceptions of residents toward ridesharing in the second (psychological shock) state, and third stages (after the psychological shock-state). The next section describes the survey design and data collection process in detail.



Fig. 2. Ridesharing introductory image used in the online survey.

3.2. Data collection

A literature review was completed on studies that analysed people's attitudes and perceptions of using shared mobility modes such as ride-hailing services (e.g. Uber), carsharing services, and ridesharing (Akbari et al., 2021; Cheah, Shimul, Liang, & Phau, 2022; Rex, Lobo, & Leckie, 2015; Wang, Wang, Wang, Wei, & Wang, 2020). Psychological questions and factors in their questionnaires were extracted and modified based on the research aim (i.e., determining the factors that influence attitudes and intention to rideshare during a period of war). The questionnaire was designed for an online platform and included socio-demographic questions (age, gender, income, car ownership, etc.), current and pre-war travel behaviours (travel mode, trip purpose, trip frequency), and Likert based psychometric questions (attitudes, current use, intention to ridesharing, etc.). The online survey was shared via social media channels and university mailing lists. Fig. 2 shows the illustration used to introduce the ridesharing concept before asking the psychometric questions. The data collection period lasted around two months and, after cleaning the missing responses, there were 318 valid responses. This is a notable data collection result considering that it occurred during a time of war.

The aims of this research project were not to determine the demographic, travel behaviours, and other characteristics of the populations in Odessa and Kyiv per se. The aim was to understand how different demographics responded to the war through their use of ridesharing and whether their propensity to do so was shaped by the same or different causal mechanisms. As such, the sampling framework sought to achieve diversity within the sample, rather than a proportionally representative group. Were the results of this study to be used to estimate population wide behaviours, sample weightings may need to be applied, but for the exploration of causality through regression techniques, this may not be required (Solon et al., 2015). Moreover, in the current war situation, gaining accurate data on the demographics of the two cities is not possible. As it has been recorded in the transport literature before (e.g., Nikitas et al., 2021) older age groups may be very hesitant to respond to an e-survey negotiating a high-tech attitude object and thus samples in many similar travel behaviour studies tend to be 'younger'. Therefore, it is likely that the sample, as shown in Table 2 is younger than that of the wider population, which is expected given the sampling strategy, with a high proportion of students (and associated student public transport pass).

There is a high degree of diversity within each of the socio-demographic variables, aiding the modelling described later in this paper. These socio-demographic characteristics are crucial for understanding the contextual factors influencing transport choices and mobility patterns among the respondents.

3.3. Method: Structural equation modelling

The research methodology consists of the following steps:

- Step 1: Internal Reliability and Validity Testing of survey responses

In the first step of this research, an evaluation of the internal reliability and validity of responses was gathered through a survey. To

Table 2
Sample characteristics.

Age	Frequency	%	Income (hryvnia)	Frequency	%
18–25	148	46.5 %	Less than 3000	10	3.1 %
26–35	66	20.8 %	3000—6000	22	6.9 %
36–45	49	15.4 %	6001—15000	79	24.8 %
46–55	34	10.7 %	15001—23000	68	21.4 %
56–65	17	5.3 %	More than 23,000	87	27.4 %
65+	4	1.3 %	Prefer not to say	52	16.4 %
Gender	**Frequency**	**%**	**Residential Area**	**Frequency**	**%**
Female	170	53.5 %	Other	4	1.3 %
Male	140	44.0 %	Rural	17	5.3 %
Non-binary	3	0.9 %	Suburban	45	14.2 %
Other	1	0.3 %	Urban	252	79.2 %
Prefer not to say	4	1.3 %	**Driving Licence**	**Frequency**	**%**
Impairment	**Frequency**	**%**	No	156	49.1 %
No	274	86.2 %	Yes	162	50.9 %
Yes	44	13.8 %	**Car ownership**	**Frequency**	**%**
Employment	**Frequency**	**%**	Without access to / own a car	180	56.6 %
Full-time	146	45.9 %	At least have access/own a car	138	43.4 %
Part-time	26	8.2 %	**Public Transport Pass**	**Frequency**	**%**
Self-employed	24	7.5 %	Disabled Person_Pass	6	1.9 %
Student	83	26.1 %	Elderly_Pass	10	3.1 %
Retired	9	2.8 %	Student_Pass	65	20.4 %
Other	27	8.5 %	Other	27	8.5 %
Unemployed	3	0.9 %	I don't have any passes	210	66.0 %
Work from Home	**Frequency**	**%**	**Subscription to Shared Mobility Modes**	**Frequency**	**%**
No	301	94.7 %	At least one shared mode	277	87.1 %
Yes	17	5.3 %	No	41	12.9 %

assess the reliability and validity of the survey instrument, statistical techniques and software, namely the IBM SPSS Statistics toolkit, were employed. This initial step is crucial to ensure the quality and accuracy of the data collected.

- Step 2: Defining the dimensions of the questions

This step involved conducting an Exploratory Factor Analysis (EFA) utilising Principal Component Analysis (PCA) in IBM SPSS Statistics. EFA is a statistical technique employed to uncover the underlying structure within a set of psychometric questions. In this study, the EFA aimed to group the psychometric questions into ten distinct factors, namely: Attitudes, Perceived Behavioural Control (PBC), Subjective Norms, Ease of Use, Usefulness, Moral Norm, Trust, Perceived Risk/Safety, Emotion, Discrimination.

- Step 3: Developing the hypothetical model to extend the Theory of Planned Behaviour

The Theory of Planned Behaviour is employed in the development of behavioural surveys to collect pertinent information about attitudes and behaviour from participants (Ajzen, 1991). This theory determines the impact of specific social and personal influencers (intentions-behavioural control) on actual behavioural changes. According to Ajzen (1991, 2020), an individual's intention is influenced by three significant factors, as illustrated in Fig. 3; *Attitudes*: these represent an individual's overall evaluation of their behaviour, wherein the individual assesses a situation or action as favourable or unfavourable. *Subjective norms*: these are the perceptions of social pressure from significant others regarding the performance of a behaviour. Subjective norms help individuals determine whether a particular activity or action is socially approved. *Perceived behavioural control*: this refers to the individual's belief in their ability to perform a specific behaviour.

Utilising the TPB, researchers (e.g., Bamberg et al., 2003; Forward, 2014; Jariyasunant et al., 2015; Schikofsky et al., 2020) investigate the pivotal role of beliefs in shaping travel behaviour. The TPB faces constraints as it does not account for other factors influencing behaviours, such as fear, threat, and past experiences. Another aspect diminishing the validity of the TPB is its assumption of perceiving behaviour as a linear decision-making process, neglecting the dynamic nature of behaviour change over time. However, to address these drawbacks, the TPB can be enhanced by incorporating different variables during the modelling phase and integrating it with other methodologies. To address the limitations of the conventional TPB, this study proposes an extended version of TPB using the factors identified through the EFA in step 2 such as moral norms, trust, perceived risk/safety, emotion and discrimination. The factors provide valuable insights into the multidimensional nature of the travel behaviour and mode preferences of individuals, allowing the extension of the TPB framework. Fig. 4 shows the conceptual model of the proposed TPB extension that incorporates the TAM and the factors discussed above to capture the intention for ridesharing during war.

There is some evidence in existing literature regarding the relationships proposed in Fig. 4, but those were not in war situations. For example, Bachmann et al. (2018) highlighted that perceived behavioural control and normative aspects strongly predict ridesharing intentions and emotional factors and moral norms influence the perception of ridesharing as a socially responsible or ethical choice, thereby affecting subjective norms and intentions. Similarly, König and Grippenkoven (2020) found that attitudes and social norms significantly impact the intention for ridesharing. In addition, Amirkiaee and Evangelopoulos (2018) confirmed a correlation between trust and actual ridesharing behaviour, especially during high-stress situations. Vaclavik et al. (2020) found that a seamless and user-friendly platform increases perceived usefulness by reducing cognitive load and improving the overall user experience. Similarly, Zhang and Liu (2022) noted that during the COVID-19 pandemic, users valued intuitive platforms, which enhanced their perception of

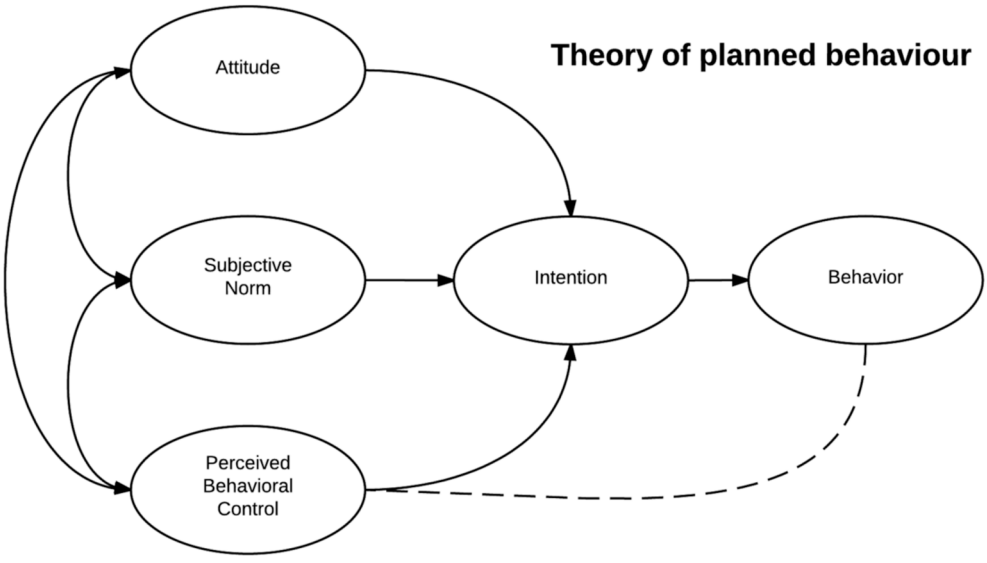


Fig. 3. Theory of Planned Behaviour (Ajzen, 1991).

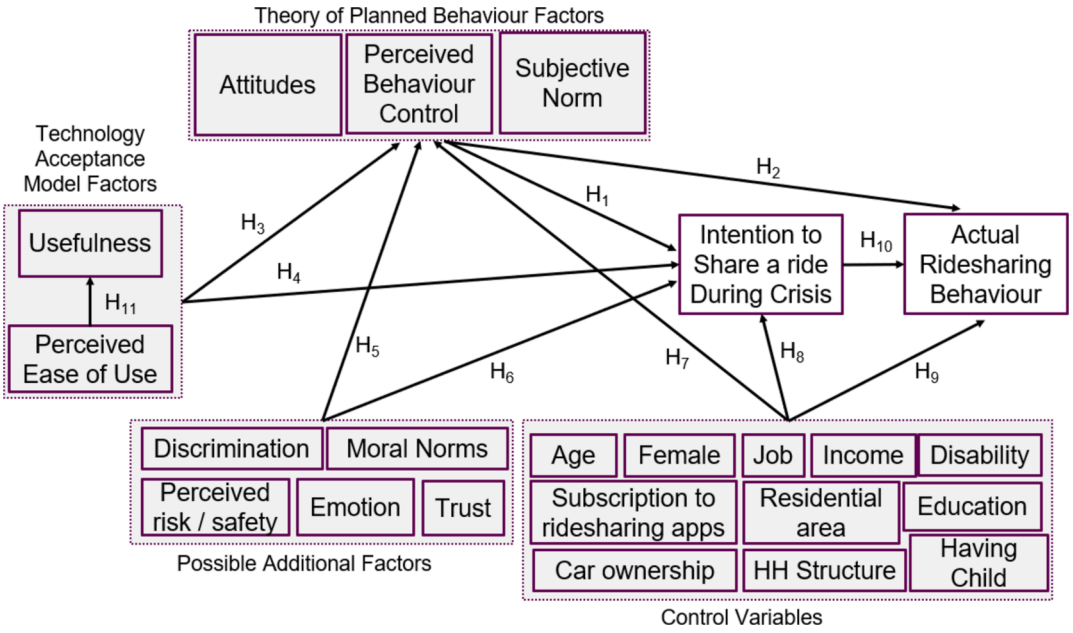


Fig. 4. Conceptual model of the extended TPB incorporating TAM, and other potential factors posited by this study.

usefulness in crises.

In this study for ridesharing during the war, the following hypotheses were proposed and tested. Some of these hypotheses were designed to test the relationship between factors in TPB:

H_1: Attitudes, perceived behavioural control, and subjective norms significantly influence ridesharing intentions.
H_2: Attitudes, perceived behavioural control, and subjective norms significantly influence current ridesharing behaviour.
H_{10} Intention to ridesharing influences current ridesharing.
Some of these hypotheses were designed to test the relationship between the factors in TAM or between TAM and TPB. For example:
H_{11} Perceived ease of use positively influences perceived usefulness.
H_3: Perceived usefulness and ease of use positively effects attitudes and perceived behavioural control towards ridesharing.
H_4: Perceived usefulness and ease of use positively influence ridesharing intentions.

Some other hypotheses (H_5, H_6) were designed to test relationships between the possible additional factors such as trust, perceived risk/safety, emotions, moral norms, and discrimination, and the core constructs of the TPB. These hypotheses aimed to explore whether these additional psychological factors had any significant influence on the traditional elements of TPB in the context of the war such as attitudes, subjective norms, perceived behavioural control, and intentions to use ridesharing. Lastly, hypotheses H_7, H_8, and H_9 were designed to test whether there is a significant relationship between control variables (age, gender, disability, income car ownership, subscription to ridesharing apps, etc.), and other factors. Other research also proof that there is relationship between these control variables and other factors (Rahimi et al., 2021, and Friman et al., (2020).

- Step 4. Analytical method choice and implementation of the analysis (CB-SEM/PLS-SEM):

Initially, two analytical methods, namely Covariance-Based Structural Equation Modelling (CB-SEM) and Partial Least Square Structural Equation Modelling (PLS-SEM) have been reviewed and examined to select appropriate methods (Dash and Paul, 2021; Memon et al., 2021).

Structural equation modelling (SEM) was created to overcome the constraints of conventional regression models, allowing for the exploration of complex causal pathways. SEM is a multivariate approach that integrates regression, factor analysis, and analysis of variance to concurrently estimate interrelated relationships among dependent variables (Fornell and Larcker, 1981). The fundamental concept of SEM is based on the work of Sewall Wright from the 1920 s. Wright sought to measure the direct causal influence between variables using path diagrams and statistical correlations. The path analysis technique is based on the correlations present within a system (Wright, 1921). More generally, the structural equation modelling approach has been extended to covariance-based SEM (CB-SEM) and variance-based SEM (PLS-SEM), which are the two predominant analytical methods within the SEM framework, each with its unique strengths and weaknesses.

Based on the criteria outlined in Table 3., CBSEM was selected as the most appropriate analytical method for this research. A key factor was having a sufficiently large sample size (n = 318) that meets the requirements of CB-SEM (Hair et al., 2017). Additionally, the

data exhibited reasonable normality, with skewness and kurtosis values within recommended ranges, avoiding issues that may arise in CB-SEM with highly non-normal data (Astrachan et al., 2014).

Further supporting the use of CB-SEM is the research objective of theory testing and confirmation, which aligns well with the strengths of CB-SEM, whereas partial least squares SEM (PLS-SEM) tends to be preferred when exploratory research and theory development are the goals (Hair et al., 2017). Since this research involves extending the established TPB, CB-SEM is well-suited for evaluating and confirming the theoretical model. IBM AMOS 28 has been selected as the analytical tool.

3.3.1. Questionnaire design

To ensure comprehensive coverage of the Theory of Planned Behaviour and Technology Acceptance Model, the questionnaire was designed to include well-established latent constructs from these models. The questions were adapted from existing literature to align with the unique context of the Ukrainian crisis. The specific latent constructs and corresponding categories are presented in Table 4. Given the crisis situation, additional sections were incorporated into the questionnaire, such as discrimination, risk and safety perception, and trust. These sections were introduced because individuals' behaviours often shift dramatically in crisis situations, necessitating the extension of traditional models of TAM and TPB to better capture these dynamics. Testing the validity of these components was important to assess whether they could effectively extend TAM and TPB in a crisis context.

Additionally, a travel resources and habits section was included to understand participants' pre-war travel behaviours. For instance, questions such as, *"What was your main mode of transport before the war for daily travel?"* and *"How frequently did you travel by this mode before the war?"* were asked to capture baseline travel habits prior to the crisis. Table 4 provides examples of these questions and their sources, which served as the basis for the constructs in the structural equation model.

To ensure the reliability of the set of statements related to these attitudinal indicators, Cronbach's alpha was measured prior to conducting factor analysis and structural equation modelling. The results demonstrated strong internal consistency across the constructs, with Cronbach's alpha values exceeding the acceptable threshold (≥ 0.7) for all key scales as seen in Table 4.

4. Results

4.1. Exploratory factor analysis (EFA)

To validate the factor structure of the measurement model, an EFA using Principal Component Analysis (PCA) for factor extraction and Direct Oblimin rotation was conducted. This approach was selected as correlations were expected between constructs, such as attitudes, trust, perceived behavioural control, and usefulness, based on the theoretical foundations of the TPB and the TAM.

The adequacy of the sample for factor analysis with a Kaiser-Meyer-Olkin (KMO) measure of 0.868, which indicates a high level of suitability for factor analysis was confirmed. Bartlett's test of sphericity was significant ($\chi 2 = 5483$, $p < 0.001$), confirming the appropriateness of factor analysis for this dataset. The EFA revealed eight components, accounting for 71.35 % of the total variance. Key constructs such as attitude, perceived behavioural control, trust, and moral norms exhibited strong loadings on their respective factors, confirming the validity of the measurement scales in the specific context of this study. The communalities and factor loadings are detailed in Table 5.

4.2. General travel behaviour before the war and during the ongoing war

This study reports the changes in transport modes when comparing the before the war to the current situation i.e., ongoing war period. Walking and cycling have seen a significant increase of 46 %, indicating a shift towards these modes, possibly due to perceived

Table 3
Guidelines for selecting the suitable analytical method in this study.

Criteria	Evidence	Evaluation	Decision
Sample Size	CB-SEM demands a lot from the data, whereas PLS-SEM is quite flexible in this aspect (Astrachan et al., 2014) CB-SEM demands a lot from data, whereas PLS-SEM is quite flexible in this aspect (Astrachan et al., 2014) PLS-SEM is more flexible but requires a minimum sample size of 30–100 (Hair et al., 2017).	The sample size in this research is 318, which is not small.	CB-SEM
Distribution of the data	CB-SEM requires the data must be normal while PLS-SEM is not (Astrachan et al., 2014). To define whether the data is normally distributed or not can be checked by the value of skewness and kurtosis. Typically, skewness and kurtosis values falling within the range of −2 to + 2 are considered acceptable for demonstrating a normal, univariate distribution (George & Mallery, 2010). Hair et al. (2010) and Bryne (2010) have also suggested that data can be considered normal when skewness ranges from −2 to + 2 and kurtosis ranges from −7 to + 7.	The skewness and kurtosis values are in the acceptable range for normality assumption.	CB-SEM
Type of modelling	If the research aims to focus on prediction and theory development, the suitable approach is PLS-SEM. Conversely, if the research objective centres on theory testing and confirmation, then the recommended method is CB-SEM (Hair et al., 2017).	The model in this research is an extension of a widely used theory called TPB.	CB-SEM

Table 4
Examples of the questions.

Section	Source	Example Question	Cronbach's Alpha
Attitude Towards Ridesharing	Cheah et al., 2022	Ridesharing makes me feel good.	0.87
Subjective Norms	(Rex et al., 2015)	During a crisis, my close friends think I should commute by ride-sharing / carpooling	0.86
Perceived Behavioural Control	(Rex et al., 2015)	During a crisis, I have full control over whether or not I adopt a ridesharing/carpooling behaviour.	0.81
Perceived Ease of Use	Akbari et al., 2021; Wang et al., 2020	If I wanted to use ride-sharing services during a crisis, it would be easy for me.	0.83
Perceived Usefulness		Using ride-sharing services would enable me to get to my destination more quickly.	0.76
Risk / Safety Perceptions	Featherman & Pavlou, 2003	Ride-sharing would put me at potential risk of physical harm due to a crisis.	0.74
Trust	Featherman & Pavlou, 2003, (Lee, Jolles, & Krabbendam, 2016)	During a crisis, it would be risky to disclose my personal information to a ridesharing platform.	0.71
Moral Norms	Culiberg, Cho, Kos Koklic, & Zabkar, 2023	During a crisis, I feel morally responsible for sharing a ride with other people.	0.70
Emotions	Dedema & Zhang, 2019	I worry about sharing a ride with others during a crisis.	0.71
Discrimination	Middleton & Zhao, 2020	Sharing a ride with a passenger of a different ethnicity could make me uncomfortable.	0.81

Table 5
Indicators' loading factors obtained from EFA.

Pattern Matrix[a]

	Component 1	2	3	4	5	6	7	8	9	10
ATT_3	0.842									
ATT_1	0.769									
ATT_6	0.761									
ATT_2	0.707									
Dis_3		0.821								
Dis_1		0.798								
Dis_4		0.768								
Dis_2		0.744								
Trust_2			0.873							
Trust_1			0.859							
Emo_2			0.483							
PBC_1				0.818						
PBC_2				0.803						
PBC_5				0.634						
PBC_3				0.627						
INT_3					−0.927					
INT_1					−0.899					
INT_2					−0.882					
INT_5					−0.842					
MorN_1						0.811				
MorN_3						0.710				
MorN_4						0.670				
SubN_1							−0.896			
SubN_2							−0.876			
SubN_3							−0.854			
Usef_1								0.833		
Usef_2								0.777		
Usef_4								0.660		
Usef_3								0.381		
Risk_2									0.835	
Risk_1									0.768	
EoU_2										0.814
EoU_3										0.807
EoU_4										0.800

Extraction Method: Principal Component Analysis.
Rotation Method: Oblimin with Kaiser Normalization.
a. Rotation converged in 12 iterations.

safety. Notably, the use of scooters for individuals with reduced mobility has emerged as a new mode. The use of taxis, including Uklon, Taxi 838, Bolt, and Uber, have increased by 47 %, possibly reflecting a preference for hired services perceived as safer. Fig. 5 illustrates these results.

Conversely, there is a substantial decline in marshrutka (van taxi) usage by 61 %, potentially due to safety concerns. Public transport, such as buses (−21 %), metro (−40 %), and trams (–23 %), has experienced decreases, suggesting a shift away from these modes. Trolleybus usage increased by 13 %, possibly considered a relatively safer public transport option. Personal vehicle usage, both alone (+11 %) and with others (+33 %), indicates a growing reliance on private transport, possibly due to perceived safety and control. Ridesharing has become more prevalent, striking a balance between safety and convenience. Overall, the shifts in transport modes reflect a nuanced response to safety considerations, convenience, and accessibility in the war context. Fig. 6 shows the frequency of shared ride experiences categorised by the companionship of the ride, distinguishing between those shared with family or friends and those shared with others.

As seen in Fig. 6, ridesharing with family and friends are more desired than ridesharing with others (other people that they do not know). Around 30 % of participants are sharing rides with family and friends at least once a week, while only around 7 % are sharing ride with others at least once a week. Some participants never ever used ridesharing neither with family/friends (around 31 % of participants), nor with other people (around 60 % of participants).

4.3. *Intention to use ridesharing during the war*

The path analysis shows significant results; a chi-square value of 697.02 with 459 degrees of freedom and a probability level of 0.000. The ratio of chi-square to degrees of freedom (CMIN/DF) was 1.519, indicating a reasonably good fit. Other measures such as the incremental fit index (IFI), and the Tucker-Lewis index (TLI) were 0.950, and 0.942, both of which were close to the desired value of 1, suggesting a good fit between the proposed model and the observed data. The root mean square error of approximation (RMSEA) was 0.041, with a 95 % confidence interval ranging from 0.043 to 0.06, indicating a close fit. The probability close (PCLOSE) was 0.351, suggesting that the model fits the data well. The Variance Inflation Factor (VIF) values reported in Table 6 demonstrate that multicollinearity is not a significant issue in the proposed model. Overall, these results demonstrate that the proposed SEM model provides a satisfactory explanation of the relationships between the variables under investigation.

Several factors, including perceived risk/safety, emotion, and discrimination, were excluded from the model due to their lack of statistically significant relationships. Notably, Trust came out as a significant factor, and it appears to partially include elements of both perceived risk/safety and emotion in its influence. However, it is worth noting that when there are no considerable demographic or ethnic differences within the sample population, discrimination may not exhibit significant effects. This suggests that discrimination's impact may be dependent on the presence of specific demographic or ethnic variations within the dataset. Another explanation is that in a war situation, discrimination may not be an identifying factor. Fig. 7 demonstrates the final form of the CB-SEM analysis using IBM

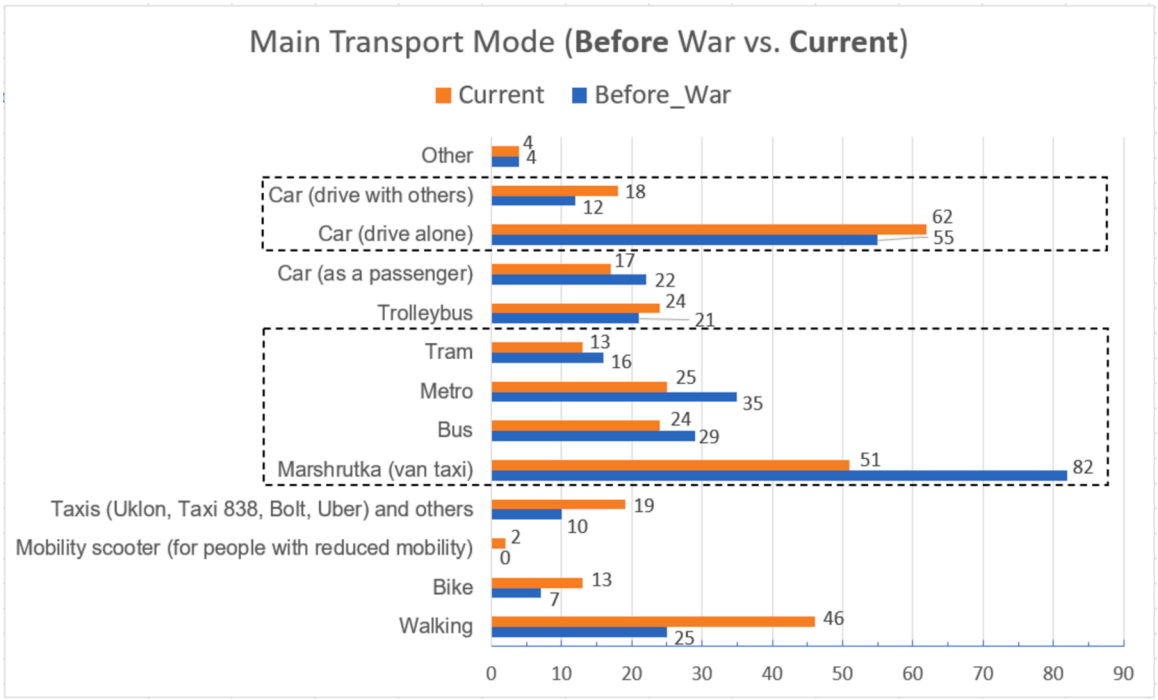


Fig. 5. Most frequent transport modes before the war and currently used.

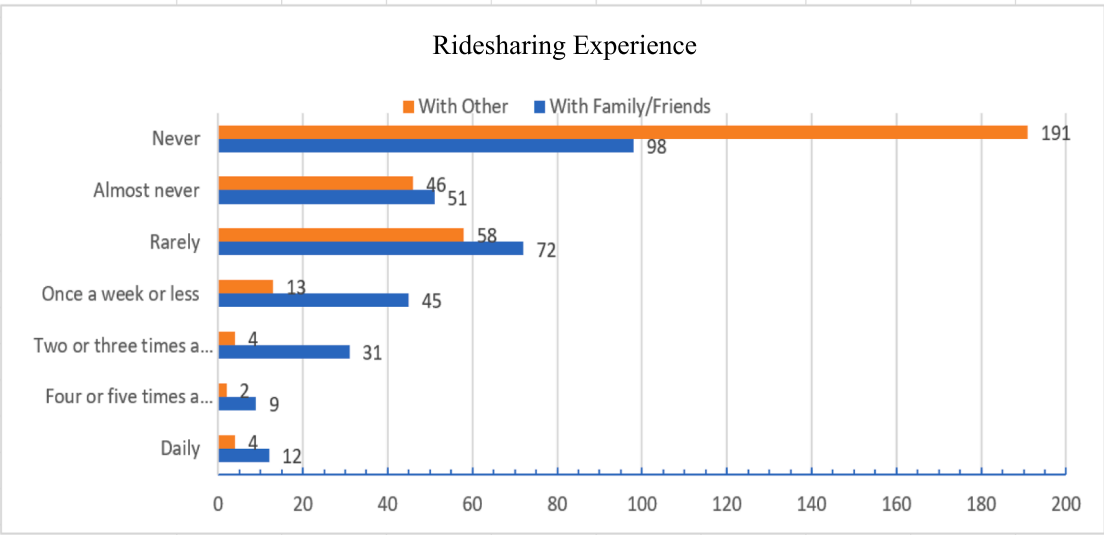


Fig. 6. Ridesharing experience among survey participants.

Table 6
Variance Inflation Factor values.

Dependent Variable	Direction	Independent Variable	Variance Inflation Factor (VIF)
Intention to Ridesharing	<—	PBC	1.194
	<—	Moral Norm	1.263
	<—	Subjective Norm	1.475
	<—	Subscription to ridesharing Apps	1.025
	<—	Attitudes	1.583
Attitudes	<—	Usefulness	1.002
	<—	Trust	1.002
Current Use	<—	Attitudes	1.356
(Ridesharing past experience)	<—	Intention to rideshare	1.351
	<—	Car ownership	1.012
PBC	<—	Car ownership	1.012
(Perceived Behavioural Control)	<—	Usefulness	1.175
	<—	Ease of Use	1.188

AMOS (Arbuckle, 2011).

The conceptual model (Fig. 4) was based on theoretical expectations, including the influence of perceived ease of use on perceived usefulness and the role of trust in affecting other variables. However, during SEM, only statistically significant relationships were retained in the final model (Fig. 7). Paths that were not supported by the data, such as the direct effect of perceived ease of use on perceived usefulness and the influence of trust, were excluded to ensure that the final model accurately reflects the data-driven findings.

The SEM results demonstrate the value of extending the TPB to explain ridesharing adoption intentions and use. While the TPB components of attitudes, PBC, and subjective norm perform as expected, the model integrates additional predictors that significantly improve its explanatory power. Table 7 demonstrates the regression estimates while all results are significant at 0.05 level, and most of them are under 0.001 level. It unveils significant connections influencing trust, attitudes, and intentions in the context of ridesharing. Specifically, the positive link between being female and trust, denoted by an estimate of 0.273 and a noteworthy p-value of 0.009, implies gender-related nuances in trust dynamics. Conversely, attitudes demonstrate a negative connection with trust, as indicated by an estimate of −0.261 and a significant p-value of 0.005, prompting a deeper exploration of the complex interplay between individual attitudes and trust within ridesharing services.

As seen in Table 7, the results of the path analysis provide support for the proposed hypotheses. PBC emerges as a pivotal factor, exhibiting positive associations with both ease of use (estimate: 0.452, p-value: ***) and perceived usefulness (estimate: 0.550, p-value: ***). These results underscore the crucial role of users' perceived control in both the adoption and perceived utility of ridesharing. Intentions to use ridesharing showcase positive links with various factors. PBC, with an estimate of 0.222 and a p-value of 0.019, along with moral norm (estimate: 0.371, p-value: ***), subjective norm (estimate: 0.172, p-value: 0.004), subscription to ridesharing apps (estimate: 0.259, p-value: 0.032), and attitudes (estimate: 0.315, p-value: ***), collectively contribute to the multifaceted nature of decision-making. Current ridesharing usage demonstrates positive associations with intention (estimate: 0.230, p-value: 0.007), car ownership (estimate: 1.336, p-value: ***), and attitudes (estimate: 0.410, p-value: ***), indicating a reinforcing

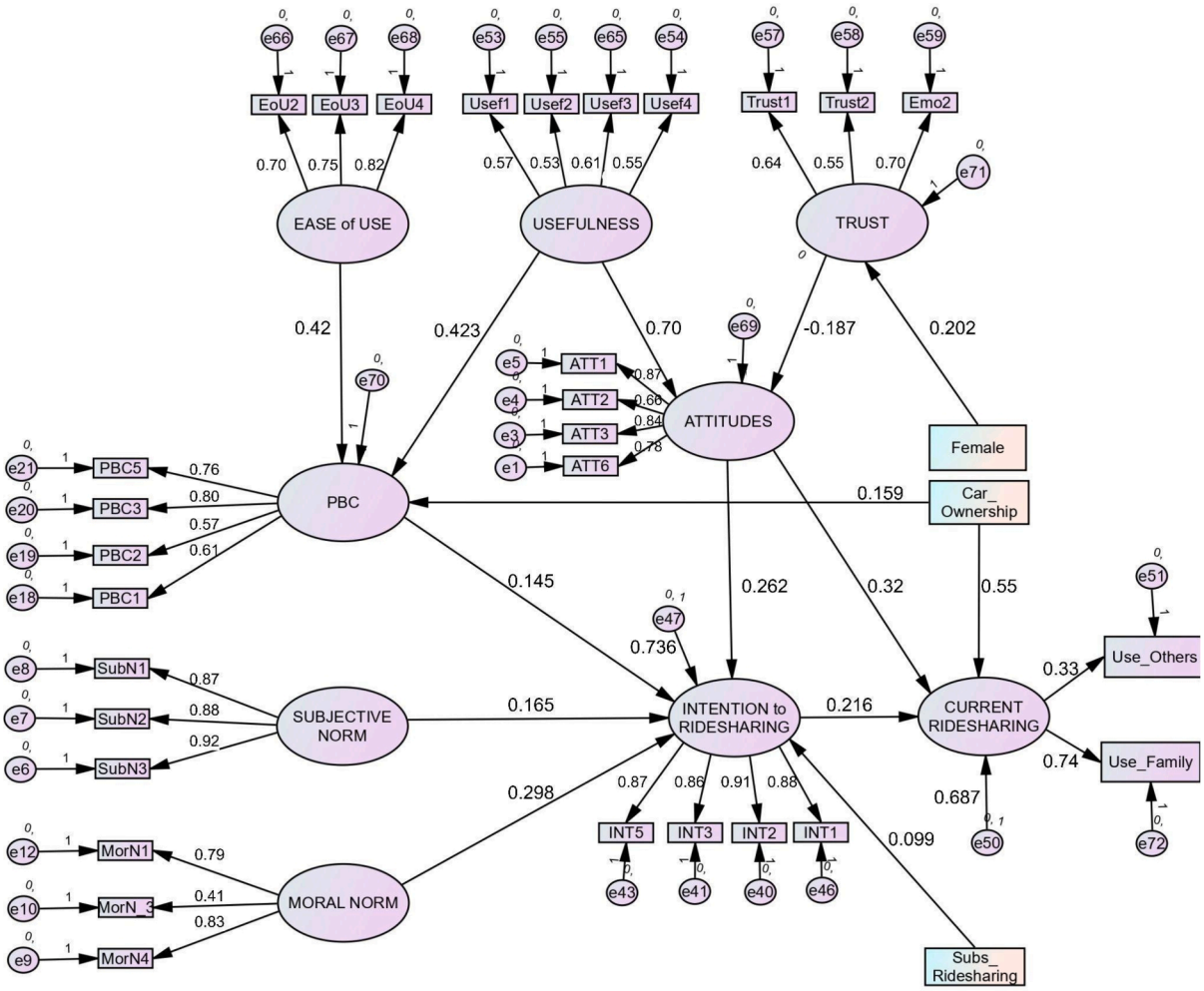


Fig. 7. Path analysis based on CB-SEM created in AMOS (.
Source: authors)

Table 7
Regression estimates.

Dependent Variable	Direction	Independent Variable	Estimate	S.E.	C.R.	P Value	Skewness/ Kurtosis[1]
Trust	<—	Female	0.273	0.105	2.609	0.009	−0.37 / −0.082
Attitudes	<—	Trust	−0.261	0.094	−2.787	0.005	−0.21 / −0.31
	<—	Usefulness	1.158	0.14	8.298	***	
PBC	<—	Ease of Use	0.452	0.087	5.194	***	−0.55/ −0.01
	<—	Usefulness	0.550	0.11	4.999	***	
	<—	Car Ownership	0.237	0.078	3.052	0.002	
Intention to Ridesharing	<—	PBC	0.222	0.095	2.342	0.019	−0.57 / −0.57
	<—	Moral Norm	0.371	0.086	4.311	***	
	<—	Subjective Norm	0.172	0.059	2.909	0.004	
	<—	Subscription to Ridesharing Apps	0.259	0.121	2.145	0.032	
	<—	Attitudes	0.315	0.075	4.179	***	
Current Use(Ridesharing past experience)	<—	Intention to Ridesharing	0.230	0.086	2.677	0.007	1.12 / 1.84
	<—	Car ownership	1.336	0.155	8.593	***	
	<—	Attitudes	0.410	0.106	3.861	***	

Note: *** means P-Value is less than 0.001 (highly significant at a 99% confidence interval).
Note: 1 means Skewness and Kurtosis values belong to dependent variables.

cycle. These nuanced insights into the complex factors influencing trust, attitudes, and intentions within the ridesharing context offer valuable information for the formulation of strategies aimed at promoting the adoption of sustainable transport alternatives.

Moreover, Table 8 presents standardised regression weights. Notably, perceived usefulness and ease of use from TAM (Technology Acceptance Model) are included in the model (perceived usefulness has a standardised regression weight of 0.70 on attitudes) and greater perceived control (perceived ease of use has a standardised regression weight of 0.42 on PBC and perceived usefulness has a weight of 0.42 on PBC). The addition of moral norms also accounts for a substantial proportion of variance in intentions beyond the TPB constructs (moral norms have a standardised regression weight of 0.30 on intentions, the third largest effect after the error term and attitudes). Further, trust is incorporated as an external variable influencing attitudes (trust has a standardised regression weight of −0.19 on attitudes). Gender demonstrated a significant and positive effect on trust, but other demographic variables did not have any significant connections. So, under war conditions, gender seems to be an important component of ridesharing behaviour by indirectly affecting trust, attitudes and intention. Lastly, car ownership is modelled as a key characteristic impacting current ridesharing use (car ownership has a standardised regression weight of 0.55 on current use, the second strongest effect). Together, these extensions of the original TPB demonstrate the utility of integrating factors from related theoretical models.

5. Discussion

This research makes several important contributions to the limited body of knowledge on ridesharing behaviour during crises that have the capacity to disrupt the functionality of urban transport systems and to affect the psychology of residents. While ridesharing has become increasingly vital for sustaining transport resilience during crises (Li et al., 2022), attitudes and behaviours towards ridesharing in these situations have not been explored in the literature. As Wang et al. (2024) suggests usage-oriented resilience (and also recovery) studies of ridesharing have been scarce. This work helps in addressing this gap by investigating factors influencing ridesharing intentions and use during the war in Ukraine, developing and validating an extended TPB model.

The findings demonstrate that extending the TPB framework with complementary constructs from related theoretical models enhances its explanatory power. The integration of perceived usefulness and perceived ease of use from the TAM proved significant, with perceived usefulness strongly predicting more positive attitudes and perceived ease of use affecting perceived behavioural control over ridesharing. This indicates the importance of emphasising the benefits of ridesharing and simplifying ridesharing systems to encourage adoption. Similar findings for peaceful times have been reported also in Wang et al., (2020)

Additionally, moral norms, used also as an extension of TPB in this new approach, was found to affect intentions to rideshare during crisis. This highlights the potential of public messaging and social awareness campaigns focused on moral norms to motivate ridesharing for the social good during disruptions. These campaigns are a useful toolkit for other transport policy interventions (Nikitas, 2018) so there is a blueprint for their use. The significant influence of trust on attitudes also highlights the need to build confidence about the reliability of ridesharing platforms among prospective users, especially for women who may feel more vulnerable or have security issues. Technological systems such as ridesharing applications should have enough safety features to make all vulnerable groups of users (including females and older people) feel safe, as also reported by Acheampong (2021) for digital mobility systems and for Sochor and Nikitas (2016) for pricing technologies during times of peace. Also, targeted policies and promotion strategies to increase the involvement of women in particular could expand the ridesharing appeal and their usage rates. Overall, the extended TPB model significantly advanced the understanding of ridesharing adoption, providing multidimensional insights.

It should be noted that during war or when natural disasters such as earthquakes occur, there might be damages and disruptions in the IT and communication infrastructures (telephone lines and internet). This can cause difficulties in using smartphone app-based mobility services such as ridesharing for arranging a shared ride (offering a shared ride or taking a shared ride). In these cases, governments can encourage people to use the predefined park and ride or park and pool sites, and mobility hubs. The effectiveness of the proposed solution and people's attitudes and intentions to use these potential locations for ridesharing (both drivers and passengers) during the crises can be further studied.

We acknowledge that the present research has some limitations that could be addressed in future studies. Our primary data, as any data referring to a self-reported e-survey may suffer to some degree by social desirability, optimism, and volunteer biases (Nikitas et al., 2021). The sample size of 318 participants, while sufficient for the analysis conducted and outstanding for wartime, was somewhat small and concentrated in two major cities in Ukraine. Larger national samples and data from diverse geographic regions could improve generalisability. The distribution of the sample from a socio-demographic characteristics perspective may not reflect the actual distribution in Ukraine. Due to the crisis situation where people are moving from Ukraine to different countries as refugees, there is uncertainty about the demographic characteristics in many instances. Comparative studies in other countries facing crises affecting mobility would also be valuable. This survey mainly focusses on the during- and after- phycological shock stages, but it does not include the long-term travel behaviour change due to the war. Policy-makers may wish to know whether the change of behaviour is temporary and long-lasting. Therefore, longitudinal data observing changes over time could provide a deeper understanding.

6. Conclusions

This exploratory study makes a significant contribution by explaining the factors shaping ridesharing intentions and use during a major crisis such as the Ukrainian war. Whilst wars are thankfully rare events and each is unique, that does not mean to say that lessons cannot be learnt from this case study which would be transferable to other conflict scenarios and other types of crises. Further work would be needed to generate statistically generalisable truths, although given the rare and unique nature of crises, such analyses would be highly problematic.

Table 8
Standardised regression weights.

Dependant Variable	Direction	Independent Variable	Estimate
Trust	<—	Female	0.202
Attitudes	<—	Trust	−0.187
PBC	<—	Ease of Use	0.420
PBC	<—	Usefulness	0.423
Attitudes	<—	Usefulness	0.700
PBC	<—	Car Ownership	0.159
Intention to Ridesharing	<—	PBC	0.145
Intention to Ridesharing	<—	Moral Norm	0.298
Intention to Ridesharing	<—	Subjective Norm	0.165
Intention to Ridesharing	<—	Subscription to Ridesharing Apps	0.099
Intention to Ridesharing	<—	e47	0.736
Intention to Ridesharing	<—	Attitudes	0.262
Current Use	<—	Intention to Ridesharing	0.216
Current Use	<—	e50	0.687
Current Use	<—	Car Ownership	0.550
Current Use	<—	Attitudes	0.320

Instead, it is suggested here to consider transferability of the results in very specific ways. Firstly, there are clear ramifications for Ukraine in their current conflict. The proposed new model offers a lens through which different approaches could be explored to better manage mobility in the current conflict. Decision-makers in Ukraine may wish to promote particular ridesharing platforms to increase public trust in these in a particular city or region prior to their being an emergency need (evacuations or major disruptions). Such promotion could target particular demographics and particular concerns as highlighted in this research. The findings reported can inform policies and initiatives to promote ridesharing as a means of improving transport resilience when systems are under such situations.

Promoting positive attitudes, enhancing perceptions of usefulness and ease of use, propagating moral norms, building trust in platforms, and encouraging women's participation appear promising for increasing reliance on ridesharing when transport capacity is disrupted. This perhaps establishes ridesharing as a critical element in improving mobility resilience during crises. However, it is notable that most ridesharing platforms and apps are usually outside of the control or influence of national or regional governments. Whereas for public transport, the schedules and routes can and were changed by governments in both Kyiv and Odessa, to minimise risks to travellers and to match the curfews that were brought in, no such control exists over the ridesharing systems.

The flexibility of such systems to function without governmental oversight is one of the benefits, but in times of crisis when governments need to direct movements of people, such as through evacuations or directing travellers along safer routes of travel, or freeing routes up for relief and military purposes, the lack of control could be a hindrance. Moreover, with increasing use of cyber-warfare the security and resilience of ridesharing apps could pose a security threat, rather than a solution to the crisis. As such, whilst it is recommended here that governments, relief agencies and other bodies could promote the use of ridesharing platforms in periods of crisis, it may be prudent for such organisations to work with the app-providers in advance of such crises. This could allow all those involved to collectively shape travel behaviours in ways that meet the user needs and could ensure the cyber-security of the platforms that would be used during the crisis.

CRediT authorship contribution statement

Nima Dadashzadeh: Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Natalia Volkova:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Data curation, Conceptualization. **Mustafa Ekmekci:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis. **Daniil Horpenko:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lee Woods:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Alexandros Nikitas:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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